2021-Ph-H1-Q22

Section: Electricity

Topic: Current, PD, Power, Resistance

A $6\,\mathrm{V}$ supply is connected across two resistors in **series**: $2\,\Omega$ and $4\,\Omega$. What is the voltage across the $4\,\Omega$ resistor?

Solution:

Voltage divides in series proportionally:

$$V = \frac{R_{\text{part}}}{R_{\text{total}}} \times V_{\text{total}} = \frac{4}{6} \times 6 = 4\,\mathrm{V}$$

Answer: B

Guidance for Students:

In series circuits, larger resistors get a larger share of the total voltage.

Revision Tip:

Voltage divider rule:

$$V_x = \frac{R_x}{R_{\text{total}}} \cdot V_{\text{total}}$$